

1. Define random variables & explain 2 types with example.

Ans. A random variable is a numerical representation of the outcome of a random experiment. It assigns a numerical value to each possible outcome in the sample space, where random experiment is a process with multiple possible outcome & the exact outcome can't be predicted beforehand.

Eg: Tossing a coin.

Types of random variables:

i). Discrete random variables: A random variable that can take on only a countable number of distinct values.

Eg: No. of heads in 3 coin flips (0, 1, 2, 3)

: No. of students absent in a class (0, 1, 2, ...).

ii). Continuous random variable: These can take on any value within a given range and the set of possible values is uncountably infinite.

Eg: Height of a person (any value between certain limit).

: Time taken to complete a task (any +ve value).

2. Let x be a number on a dice when thrown. Find mean & variance of x . Write Probability distribution function.

x :	1	2	3	4	5	6
$P(x)$:	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Ans. $\therefore$ this is a discrete probability distribution;

$$\text{Mean, } \mu = E(x) = \sum x \cdot P(x).$$

$$= (1 \times \frac{1}{6}) + 2(\frac{1}{6}) + 3(\frac{1}{6}) + 4(\frac{1}{6}) + 5(\frac{1}{6}) + 6(\frac{1}{6})$$

$$= (1+2+3+4+5+6) \frac{1}{6} = \frac{21}{6} = \underline{\underline{3.5}}$$

$$\text{Variance, } \sigma^2 = E(x^2) - [E(x)]^2.$$

$$E(x^2) = \sum x^2 p(x)$$

$$= \left[1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 \right] \frac{1}{6}$$

$$= \frac{91}{6}$$

$$\therefore \sigma^2 = \frac{91}{6} - (3.5)^2 = \frac{35}{12} = \underline{\underline{\frac{2.91}{12}}}$$

3. Given $f(x) = kx^4 e^{-x}$; $0 < x < \infty$ is a probability distribution function. Determine the value of k .

Ans. The variable is continuous here,

$\therefore$ by continuous probability distribution,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^{\infty} f(x) dx = \int_0^{\infty} kx^4 e^{-x} dx = 1 \quad \left[\because \text{from } -\infty \text{ to } 0 \text{ function has no value} \right]$$

$$= k \int_0^{\infty} x^4 e^{-x} dx = 1 \quad \text{--- (1)}$$

Gamma function:

$$\Gamma(n) = \int_0^{\infty} x^{n-1} e^{-x} dx \quad ; \quad n > 0 \quad \text{--- (2)}$$

Comparing (1) & (2);

$$n = \underline{\underline{5}}$$

$$\therefore \int_0^{\infty} f(x) dx = k \Gamma(5) = 4! \times k = 1$$

$$\text{i.e., } 24k = 1$$

$$\therefore k = \underline{\underline{\frac{1}{24}}}$$

4. Obtain the distribution function of mean of total no. of heads occurring in 3 tosses of an unbiased coin. x denotes no. of heads (0, 1, 2, 3).

Ans. No. of heads occurring in 3 tosses can be 0, 1, 2, 3.

Sample space $.S = \{HHH, HHT, HTH, THH, HTT, THT, THT, TTT\}$

Probability distribution function:

$$P(X=0) = \frac{1}{8} (TTT)$$

$$P(X=1) = \frac{1}{8} (HHT, HTH, HTT)$$

$$P(X=2) = \frac{1}{8} (THH, THT, TTH)$$

$$P(X=3) = \frac{1}{8} (HHH)$$

$\therefore x :$	0	1	2	3
$P(x) :$	$1/8$	$3/8$	$3/8$	$1/8$

$$\text{Mean, } \mu = E(X) = \sum x \cdot P(x).$$

$$= (0 \times 1/8) + 1(3/8) + (2 \times 3/8) + (3 \times 1/8)$$

$$= 1/8 + 3/8 + 6/8 + 3/8$$

$$= \underline{\underline{1.5}}$$

5. A random variable x has the following probability distribution function.

$x :$	0	1	2	3	4	5	6	7
$P(x) :$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

Find (i) k , (ii) $P(X \geq 6)$

Use $\sum P(x) = 1$ property. Get 2 values of k .

Ans $\therefore$ the sum of probabilities of a probability distribution function of random variable is 1.

$$\therefore 0 + k + 2k + 2k + 3k + k^2 + 2k^2 + (7k^2 + k) = 1$$

$$\Rightarrow 10k^2 + 9k - 1 = 0.$$

$$(10k - 1)(k + 1) = 0$$

$$\therefore k = -1, \underline{\underline{\frac{1}{10}}}.$$

$k = -1$ is not possible as the probability of an event is never negative.

$$P(x \geq 6) = P(x = 6) + P(x = 7).$$

$$= 2k^2 + 7k^2 + k.$$

Substituting $k = \frac{1}{10}$;

$$P(x \geq 6) = 2\left(\frac{1}{10}\right)^2 + 7\left(\frac{1}{10}\right)^2 + \left(\frac{1}{10}\right)$$

$$= \underline{\underline{0.19}}$$

6. If x is a poisson variant such that $P(x=2) = 9P(x=4) + 90P(x=6)$
Find mean & standard deviation.

Ans. $P(x=k) = \frac{e^{-\lambda} \lambda^k}{k!}$

$$P(x=2) = \frac{e^{-\lambda} \lambda^2}{2!} = \frac{e^{-\lambda} \lambda^2}{2}$$

$$P(x=4) = \frac{e^{-\lambda} \lambda^4}{4!} = \frac{e^{-\lambda} \lambda^4}{24}$$

$$P(x=6) = \frac{e^{-\lambda} \lambda^6}{6!} = \frac{e^{-\lambda} \lambda^6}{720}$$

Substituting;

$$\frac{e^{-\lambda} \lambda^2}{2} = 9 \cdot \frac{e^{-\lambda} \lambda^4}{24} + 90 \cdot \frac{e^{-\lambda} \lambda^6}{720}$$

Cancel $e^{-\lambda}$ from both sides assuming $\lambda > 0$.

$$\therefore \frac{\lambda^2}{2} = 9 \cdot \frac{\lambda^4}{24} + 90 \cdot \frac{\lambda^6}{720}$$

$$\Rightarrow \frac{90\lambda^6}{720} =$$

$$\frac{\lambda^2}{2} = \frac{3\lambda^4}{8} + \frac{\lambda^6}{8}$$

Multiplying through out by 8; we get;

$$4\lambda^2 = 3\lambda^4 + \lambda^6 \Rightarrow \lambda^6 + 3\lambda^4 - 4\lambda^2 = 0.$$

Mean $\Rightarrow \lambda = 1 \Rightarrow$ ~~variance, $\lambda = 1$~~

standard deviation, $\sqrt{\lambda} = \sqrt{1} = 1$

7. The probability of a Poisson variant taking the values 3 & 4 are equal. calculate the probability of the variants taking the value 0 & 1.

Ans. $P(x=3) = P(x=4)$

$$\frac{e^{-\lambda} \lambda^3}{3!} = \frac{e^{-\lambda} \lambda^4}{4!}$$

$$\therefore \lambda = \frac{4!}{3!} = \underline{4}$$

$$P(x=0) = \frac{e^{-4} \times 4^0}{0!} = \frac{1}{e^4} = \underline{0.018}$$

$$P(x=1) = \frac{e^{-4} 4^1}{1!} = \frac{4}{e^4} = \underline{0.073}$$

8. The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured find the probability that (i). exactly 2 will be defective. (ii). atleast 2 will be defective, (iii). none will be defective.

Ans. Probability of defective pen = $\frac{1}{10} = \underline{0.1}$
 Probability of non defective pen = $1 - 0.1 = \underline{0.9}$
 $n = 12$.

- i). Probability that exactly 2 will be defective

$$\rightarrow {}^{12}C_2 (0.1)^2 (0.9)^{10} = \underline{0.2301}.$$

- ii). Probability that atleast 2 will be defective

$$\begin{aligned} &\rightarrow 1 - P(\text{either none or one is non-defective}) \\ &\rightarrow 1 - [{}^{12}C_0 (0.9)^{12} + {}^{12}C_1 (0.9)^{11} (0.1)] \\ &\rightarrow \underline{0.3412}. \end{aligned}$$

- iii). Probability that none will be defective.

$$\rightarrow {}^{12}C_0 (0.9)^{12} = \underline{0.2833}.$$

9. Out of 800 families with 4 children each, how many families would be expected to have (i). 2 boys & 2 girls, (ii) atleast 1 boy, (iii). No girl, (iv). atmost 2 girls. Assume equal possibilities for boys & girls.

Ans. Probability of 2 girls = $\frac{3}{4} = \underline{\frac{1}{2}}$.

$$\text{Probability of 2 boys} = \frac{2}{4} = \underline{\frac{1}{2}}$$

- i). $\therefore$ probability of exactly 2 boys & 2 girls =

$$P(x=2) = {}^nC_x p^x q^{n-x}$$

$$= {}^4C_2 (0.5)^2 (0.5)^2 = \underline{\underline{\frac{3}{8}}}$$

$$\text{No. of families with 2 boys \& 2 girls} = 800 \times \frac{3}{8} = \underline{\underline{300}}$$

ii). atleast one boy $\Rightarrow$.

$$P(x \geq 1) = P(x=1) + P(x=2) + P(x=3) + P(x=4).$$

$$= 1 - P(x=0)$$

$$= 1 - {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4$$

$$= 1 - 0.0625 = \underline{\underline{0.9375}}$$

$$\text{For 800 families} \Rightarrow 800 \times 0.9375 = \underline{\underline{750}}$$

iii). No girl,

$$P(x=4) \Rightarrow {}^4C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 = \underline{\underline{0.0625}}$$

$$\text{No. of families} \Rightarrow 800 \times 0.0625 = \underline{\underline{50.}}$$

iv). atmost 2 girls.

$$P(x \leq 2) = P(x=0) + P(x=1) + P(x=2).$$

$$= {}^4C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 + {}^4C_1 \left(\frac{1}{2}\right) \left(\frac{1}{2}\right)^3 + \frac{3}{8}$$

$$= 0.0625 + 0.25 + \frac{3}{8}$$

$$= \underline{\underline{0.6875}}$$

$$\Rightarrow 800 \times 0.6875 = \underline{\underline{550}}$$

10. In a certain factory, turning out blades, there is a small chance that of 0.002 per any blade to be defective. The blades are supplied in packets of 10. Use poisson distribution to calculate the approx. no. of packets containing (i). No defective (ii) 1 defective

(iii) 2 defective blades in a consignment of 10,000 packets.

Ans. $n = 10$, $p = 0.002$

Binomial distribution can be approximated using Poisson with parameter, $m = np = 0.002 \times 10 = \underline{0.02}$

Let x equals to no. of defective blades in a packet.

$$P_0 = P(x=0) = e^{-0.02} = \underline{0.9802}.$$

Using the formula

$$P_{x+1} = P_x \cdot \frac{m}{x+1}$$

$$P_1 = P_0 \cdot \frac{0.02}{1} = \underline{0.019604}$$

$$P_2 = P_1 \cdot \frac{0.02}{2} = \underline{0.00019604}$$

$$P_3 = P_2 \cdot \frac{0.02}{3} \approx \underline{0.}$$

$$\therefore n_0 = 10,000 P_0 = \underline{9802}$$

$$n_1 = 10,000 P_1 = \underline{196}$$

$$n_2 = 10,000 P_2 = \underline{2}$$

$$n_3 = \underline{0.}$$

11. Derive mean & variance of normal distribution.

Ans. Let x be a random variable following a normal distribution.

$$X \sim N(\mu, \sigma^2).$$

Mean or expected value of $x \Rightarrow$

$$E(x) = \mu.$$

$$E(x) = \int_{-\infty}^{\infty} x \cdot f(x) dx.$$

$$E(x) = \int_{-\infty}^{\infty} x \frac{1}{\sqrt{2\pi} \sigma} e^{\left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right]} dx$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} x \cdot e^{\left[-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2\right]} dx$$

Substituting $z = x - \mu$, we have

$$E(x) = \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty-\mu}^{\infty-\mu} (z+\mu) e^{\left[-\frac{1}{2} \left(\frac{z}{\sigma}\right)^2\right]} \frac{d(z+\mu)}{d(z+\mu)}.$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \int_{-\infty}^{\infty} (z+\mu) e^{\left[-\frac{1}{2} \left(\frac{z}{\sigma}\right)^2\right]} dz.$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \left[\int_{-\infty}^{\infty} z \cdot \exp\left[-\frac{1}{2} \left(\frac{z}{\sigma}\right)^2\right] dz + \mu \int_{-\infty}^{\infty} e^{\left[-\frac{1}{2} \left(\frac{z}{\sigma}\right)^2\right]} dz \right]$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \left[\int_{-\infty}^{\infty} z e^{-\frac{1}{2\sigma^2} \cdot z^2} dz + \mu \int_{-\infty}^{\infty} e^{\left(-\frac{1}{2\sigma^2} \cdot z^2\right)} dz \right]$$

General antiderivatives are:

$$\int x \cdot e^{-ax^2} dx = -\frac{1}{2a} e^{-ax^2}$$

$$\int e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}} \operatorname{erf}(\sqrt{ax}).$$

where, erf is error function.

$$E(x) = \frac{1}{\sqrt{2\pi} \sigma} \left[-\sigma^2 e^{\left(-\frac{1}{2\sigma^2} \cdot z^2\right)} \right]_{-\infty}^{\infty} + \mu \left[\sqrt{\frac{\pi}{2}} \sigma \operatorname{erf}\left(\frac{1}{\sqrt{2}\sigma} z\right) \right]_{-\infty}^{\infty}$$

$$= \frac{1}{\sqrt{2\pi} \sigma} \left[\lim_{z \rightarrow \infty} \left(-\sigma^2 e^{\left(-\frac{1}{2\sigma^2} \cdot z^2\right)} \right) - \lim_{z \rightarrow -\infty} \left(-\sigma^2 e^{\left(-\frac{1}{2\sigma^2} \cdot z^2\right)} \right) \right]$$

$$+ \mu \left[\lim_{z \rightarrow \infty} \sqrt{\frac{\pi}{2}} \sigma \operatorname{erf}\left(\frac{1}{\sqrt{2}\sigma} z\right) - \lim_{z \rightarrow -\infty} \left(\sqrt{\frac{\pi}{2}} \sigma \operatorname{erf}\left(\frac{1}{\sqrt{2}\sigma} z\right) \right) \right]$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{2\pi}\sigma} (0-0) + \mu \left(\sqrt{\frac{\pi}{2}}\sigma - \left(-\sqrt{\frac{\pi}{2}}\sigma \right) \right) \\
 &= \frac{1}{\sqrt{2\pi}\sigma} \mu \cdot 2\sqrt{\frac{\pi}{2}}\sigma \\
 &= \mu.
 \end{aligned}$$

Variance.

Let x be a random variable

$$X \sim N(\mu, \sigma^2)$$

Variance of $X \Rightarrow \text{Var}(X) = \sigma^2$.

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - E(x))^2 f_X(x) dx.$$

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu)^2 \cdot \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right] dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (x - \mu)^2 \exp\left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right] dx.$$

Substituting $z = x - \mu$.

$$\text{Var}(X) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} (\sqrt{2}\sigma z)^2 \cdot \exp\left[-\frac{1}{2} \left(\frac{\sqrt{2}\sigma z}{\sigma}\right)^2\right] d(\sqrt{2}\sigma z)$$

$$= \frac{1}{\sqrt{2\pi}\sigma} 2\sigma^2 \cdot \sqrt{2}\sigma \int_{-\infty}^{\infty} z^2 \cdot \exp(-z^2) dz$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2} dz.$$

Since the integrand is symmetric with respect to $z=0$,

$$\text{Var}(X) = \frac{4\sigma^2}{\sqrt{\pi}} \int_0^{\infty} z^2 e^{-z^2} dz$$

If we define $z = x^2$ then $x = \sqrt{z}$ & $dx = \frac{1}{2} z^{-1/2} dz$

Substituting this into integral

$$\begin{aligned}\text{Var}(x) &= \frac{4\sigma^2}{\sqrt{\pi}} \int_0^{\infty} z \cdot e^{-z} \cdot \frac{1}{2} z^{-1/2} \\ &= \frac{2\sigma^2}{\sqrt{\pi}} \int_0^{\infty} z^{3/2-1} e^{-z} dz\end{aligned}$$

Using the definition of gamma function.

$$\Gamma x = \int_0^{\infty} z^{x-1} e^{-z} dz$$

$$\therefore \text{Var}(x) = \frac{2\sigma^2}{\sqrt{\pi}} \cdot \Gamma_{3/2} = \frac{2\sigma^2}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \sigma^2.$$

12. Fit an equation of the form

$y = ax^b$ to the following data.

x :	1	2	3	4	5	6
y :	2.98	4.26	5.21	6.1	6.8	7.5

Ans.

$$y = ax^b, n = 6.$$

Applying $\log_e$ on both sides,

$$\log y = \log a + b \log x$$

$$Y = A + bX$$

Substituting $Y = \log y$, $A = \log a$, $X = \log x$.

$$nA + b \sum X = \sum Y$$

$$A \sum X + b \sum X^2 = \sum XY$$

x	y	$X = \ln x$	$Y = \ln y$	X^2	XY
1	2.98	0	1.0919	0	0
2	4.26	0.6931	1.4493	0.4805	1.0045
3	5.21	1.0986	1.6506	1.2069	1.8133
4	6.1	1.3863	1.8083	1.9218	2.5068
5	6.8	1.6094	1.9169	2.5903	3.0851
6	7.5	1.7918	2.0149	3.2104	3.6103
Σ		6.5792	9.9319	9.4099	12.0201

$$6A + 6.5792b = 9.9319$$

$$6.5792A + 9.4099b = 12.0201$$

Solving this we get, $A = \underline{1.0912}$ - $b = \underline{0.5144}$.

$$\log a = 1.0912 \Rightarrow a = e^{1.0912} = 2.9778$$

$$b = 0.5144$$

$$\therefore \text{required curve} \Rightarrow y = ax^b$$

$$y = 2.9778 \times x^{0.5144}$$

13. Fit a curve of the form $Y = ae^{bx}$ to the following data.

x :	1	2	3	4	5	6
y :	1.6	4.5	13.8	40.2	125	300

Ans.

$$y = ae^{bx}$$

Taking log on both sides;

$$\log y = \log ae^{bx}$$

$$\log y = \log a + bx \log e$$

$$Y = \log y, \quad A = \log a, \quad B = b \log e$$

$$\therefore Y = A + Bx$$

normal equation $\Rightarrow$

$$\sum Y = A \sum 1 + B \sum x$$

$$\sum xY = A \sum x + B \sum x^2$$

x	y	x^2	$Y = \log y$	xY
1	1.6	1	0.204	0.204
2	4.5	4	0.653	1.306
3	13.8	9	1.140	3.42
4	40.2	16	1.604	6.416
5	125	25	2.097	10.485
6	300	36	2.477	14.862

12. r.

$$\sum x = 21, \quad \sum x^2 = 91, \quad \sum y = 8.175, \quad \sum xy = 36.694$$

$$\sum y = A \sum x + B \sum x^2$$

$$8.175 = 6A + 21B$$

$$\sum xy = A \sum x + B \sum x^2$$

$$36.694 = 21A + 91B$$

$$\text{Solving this, } B = 0.462.$$

$$A = \text{also } 0.2545$$

$$\log a = A \Rightarrow a = \underline{0.557}.$$

$$B = b \log e \Rightarrow b = \frac{B}{\log e} = \frac{0.462}{\log e} = \underline{1.074}$$

$$y = ae^{bx}$$

$$y = (0.557)e^{1.074x}$$